

Network Administration 9 days

Protocols
 Topologies
 networking Devices
 63.0.0.0 similar nws
 63.45.67.80 nw add
 63.255.255.255 ip address
 Hub b'cast add
 Switch
 Bridge
 Repeater

=====
 Router/Gateway
 Firewall
 L3, L4 Switches
 BRouter Bridge + Router

cables
 IP address & Mac Address

OSI & TCP/IP Model

=====
 IP Classes:
 A 0-127 1-126
 B 128-191
 C 192-223
 D 224-239 Multicasting (nw devices---inter-communication)
 E 240-255 R&D

Subnet Mask helps to differentiate between nw bits & host bits
 (255-->nw, 0--->host)

A 255.0.0.0
 B 255.255.0.0
 C 255.255.255.0

=====
 CIDR Classless Inter-Domain Routing
 /1 number of bits given to nw
 /8 8 bits --->nw
 255.0.0.0

Subnet Mask	CIDR Value
128.0.0.0 10000000.00000000.00000000.00000000	/1
192.0.0.0 11000000.00000000.00000000.00000000	/2
255.0.0.0 11111111.00000000.00000000.00000000	/8
255.128.0.0 11111111.10000000.00000000.00000000	/9
	/10
255.255.0.0	/16
	/22
255.255.252.0	
	/32

L3 Broadcast	255.255.255.255	(all nws & all hosts)
L2 Broadcast	ff:ff:ff:ff:ff:ff	

Subnetting networks. / Sub-Networking: Dividing one large network into multiple smaller networks. VV IMP

1. Easy Manageability
 2. Security
 3. Increases the number of Broadcast domain.
- Class-X
100

X-A	X-B
50	50

X-A	X-B	X-C	X-D
25	25	25	25

HR	Admin	Engg	Programmers
192.168.1.1	192.168.1.2		

192.168.1.0/24	NW Address
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192.168.1.1	Usable IP address
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192.168.1.254	Usable
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192.168.1.255

B'cast Add

Class C Subnetting:

borrow bits from host bits

1. 192.168.1.0/25

/25----> 8.8.8.1

Subnet Mask 255.255.255.128

n=1

 $2^1===2$

(Total Subnets)

32-25===7

host bits

 $2^7===128$

Block Range (IPs per subnet)

128-2===126

Valid IPs

(0-255)

256-128===128

(jump value)

I

192.168.1.0

Address

192.168.1.1

Ips

II

192.168.1.128

Subnet

192.168.1.129

Valid

192.168.1.126

192.168.1.127

Add

192.168.1.254

192.168.1.255

B'cast

pc1

connect

192.168.1.21

255.255.255.128

pc2

192.168.1.14

255.255.255.128

pc1

cannot connect

192.168.1.21

255.255.255.128

pc2

192.168.1.140

255.255.255.128

2. 192.168.1.0/26

/26

8.8.8.2

Subnet Mask===255.255.255.192

n=2

 $2^2===4$

Total Subnets

32-26===6

Host bits

 $2^6===64$

Block Size

64-2===62

Usable IPs

256-192===64

Jump Value

IV

I

192.168.1.0

192.168.1.192

192.168.1.1

192.168.1.193

II

192.168.1.64

192.168.1.65

III

192.168.1.128

192.168.1.129

192.168.1.62	192.168.1.126	192.168.1.190
192.168.1.254		
192.168.1.63	192.168.1.127	192.168.1.191
192.168.1.255		

=====

=====

3. 192.168.1.0/27

27

8.8.8.3

Subnet Mask====255.255.255.224

n=3

$2^3=8$

Total Subnets

32-27==5

$2^5=32$

Host bits

Block Size

32-2==30

Usable IPs

256-224==32

Jump Value

I							
192.168.1.0	32	64	96	128	160	192	224
192.168.1.1	33	65	97	129	161	193	225
192.168.1.30	62	94	126	158	190	222	254
192.168.1.31	63	95	127	159	191	223	255

=====

=====

3. 192.168.1.0/28

28

8.8.8.4

Subnet Mask====255.255.255.240

n=3

$2^4=16$

Total Subnets

32-28==4

$2^4=16$

Host bits

Block Size

16-2==14

Usable IPs

256-240==16

Jump Value

0	16	32	48	64	80	96	108	128	144	160	176	192	208
224	240												
1	17												
225	241												
14	30												
238	254												
15	31												
239	255												

=====

4. 192.168.1.0/30

Solve this

255.255.255.252

64 SN
4 BS
2 VH

256-252===4

JV

0	4	8
248	252	
1	5	
249	253	
2	6	
250	254	
3	7	
251	255	

Summary of Class C Subnetting:

CIDR Size	Subnet Mask Usable IPs	Total Subnets	Block
/25	255.255.255.128	2	128
/26	255.255.255.192	4	64
/27	255.255.255.224	8	32
/28	255.255.255.240	16	16
/29	255.255.255.248	32	8
/30	255.255.255.252	64	4

Class B Subnetting

128-191

172.16.0.0/16 default CIDR---/16

1. 172.16.0.0/18 /18---> 8.8.2.0 SM---> 255.255.192.0

n=2

2^2===4

32-18===14

2^14=====16384

16384-2 ===16382

256-192==64

Subnets

HB

BS

Valid IPs

JV

172.16.0.0	172.16.64.0	172.16.128.0	172.16.192.0
172.16.0.1	172.16.64.1	172.16.128.1	172.16.192.1

172.16.63.254	172.16.127.254	172.16.191.254	172.16.255.254
172.16.63.255	172.16.127.255	172.16.191.255	172.16.255.255

2. 172.16.0.0/20
255.255.240.0

/20---> 8.8.4.0

SM--->

	n=4	$2^4=16$ $32-20=12$ $2^{12}=4096$ $4096-2=4094$	Subnets HB BS / per subnet Valid IPs / per
subnet		256-240=16	JV

I XVI	II	XV
172.16.0.0 172.16.240.0 172.16.0.1 172.16.240.1	172.16.16.0 172.16.16.1	172.16.224.0 172.16.224.1
172.16.15.254 172.16.255.254 172.16.15.255 172.16.255.255	172.16.31.254 172.16.31.255	172.16.239.254 172.16.239.255

```
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=====
3. 172.16.0.0/24          24----> 8.8.8.0          SM---->
255.255.255.0
```

	n=8(Borrowed bits)	
	$2^8=256$ $32-24=8$ $2^8=256$ $256-2=254$	Total Subnets HB BS / per subnet Valid IPs / per
subnet	256-255=1	JV

I XVI	II	XV
172.16.0.0 172.16.255.0 172.16.0.1 172.16.255.1	172.16.1.0 172.16.1.1	172.16.254.0 172.16.254.1
172.16.0.254 172.16.255.254 172.16.0.255 172.16.255.255	172.16.1.254 172.16.1.255	172.16.254.254 172.16.254.255

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=====
=====
4. 172.16.0.0/29          29----> 8.8.8.5          SM---->
255.255.255.248
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	n=13(Borrowed bits)	
	$2^{13}=8192$ $32-29=3$ $2^3=8$	Total Subnets HB BS / per subnet

subnet	8-2====6	Valid IPs / per
	256-248==8	JV

172.16.0.0	172.16.0.8	172.16.255.240
172.16.255.248		
172.16.0.1	172.16.0.9	172.16.255.241
172.16.255.249		

172.16.0.6	172.16.0.14	172.16.255.246
172.16.255.254		
172.16.0.7	172.16.0.15	172.16.255.247
172.16.255.255		

=====
 =====
 Class A ----- /9---/30

1. 10.0.0.0/16	8.8.0.0	255.255.0.0
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n==8
 256 Subnets
 16 HB

2^16===65536
 65536-2==65534
 BS /Subnet
 Usable IP/Subnet

10.0.0.0	10.1.0.0	10.2.0.0
10.254.0.0	10.255.0.0	
10.0.0.1	10.1.0.1	10.2.0.1
10.254.0.1	10.255.0.1	

10.0.255.254	10.1.255.254	10.2.255.254
10.254.255.254	10.255.255.254	
10.0.255.255	10.1.255.255	10.2.255.255
10.254.255.255	10.255.255.255	

=====
 =====
 2. 10.0.0.0/19

	8.8.3.0	255.255.224.0
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n==11
 2048 Subnets
 13 HB

2^13===8192
 8192-2==8190
 BS /Subnet
 Usable IP/Subnet

256-224==32

10.0.0.0	10.0.32.0	10.255.192.0
10.255.224.0		
10.0.0.1	10.0.32.1	10.255.192.1
10.255.224.1		

10.0.31.254	10.0.63.254	10.255.223.254
10.255.255.254		
10.0.31.255	10.0.63.255	10.255.223.255
10.255.255.255		

=====

3. 10.0.0.0/28

8.8.8.4 255.255.255.240

n==20
10,48576
4

Subnets
HB

$2^4=16$
 $16-2=14$

BS /Subnet
Usable IP/Subnet

$256-240=16$

10.0.0.0	10.0.0.16	10.255.255.224
10.255.255.240		
10.0.0.1	10.0.0.17	10.255.255.225
10.255.255.241		

10.0.0.14	10.0.0.30	10.255.255.254
10.255.255.254		
10.0.0.15	10.0.0.31	10.255.255.255
10.255.255.255		

=====

80.160.0.0/20 8.8.4.0
255.255.240.0

12 borrowed $2^{12}=4096$

$4096-2$ 4094

=====

192.168.1.134/30 is the valid IP address, you need to find Subnet address for this?
192.168.1.132

255.255.255.252
4 JV

0	4	8	128	132	136
				133	
				134	
				135	

192.168.1.193/27 is the valid IP address, you need to find Broadcast address for this?
8.8.8.3 SM---> 255.255.255.224 32---JV

0	192	224
	193	

223

172.16.96.43/24 is the valid IP address, you need to find subnet address(172.16.96.0) & last usable IP for this(172.16.96.254)

8.8.8.0 SM--->255.255.255.0 1---JV

172.16.96.0 172.16.97.0

172.16.96.43

172.16.96.254

172.16.96.255

10.51.34.255/29 is the broadcast IP address, you need to find subnet address(10.51.34.248)& first usable IP for this(10.51.34.249)

8.8.8.5 SM---->255.255.255.248 256-248==8

10.51.34.248

10.51.34.249

10.51.34.255

=====

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Route Summarization: Combining multiple routing table entries into one entry is called Route Summarization or as Supernetting

1. Saves Router's memory space
2. Saves time
3. less bandwidth consumption

192.168.1.0/20 Supernetting

192.168.1.0/24 Networking

192.168.1.0/27 Subnetting

RouterA

RouterB

A Summary Nw(AZ)

AZ (De-summarize)

B

C

D

Z

Types of Summarization:

1. Manual NW Admin will do the route summarization
2. Auto Router's protocol will do

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R-A Summary Address--->10.0.0.0/8

R-B

10.0.0.0/16

10.1.0.0/16

10.2.0.0/16

10.3.0.0/16

10.4.0.0/16

10.255.0.0/16

10.0.0.0/8

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